Interactive Teaching Model for Pulse Oximetry

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Design Project Goal

Develop an interactive pulse oximetry teaching model tailored for Duke Ignite students and similar STEM learners

⇾ This model should promote understanding of biomedical engineering principles behind pulse oximetry

⇾ It should allow students to physically explore how light absorption and detection work when measuring blood oxygen levels

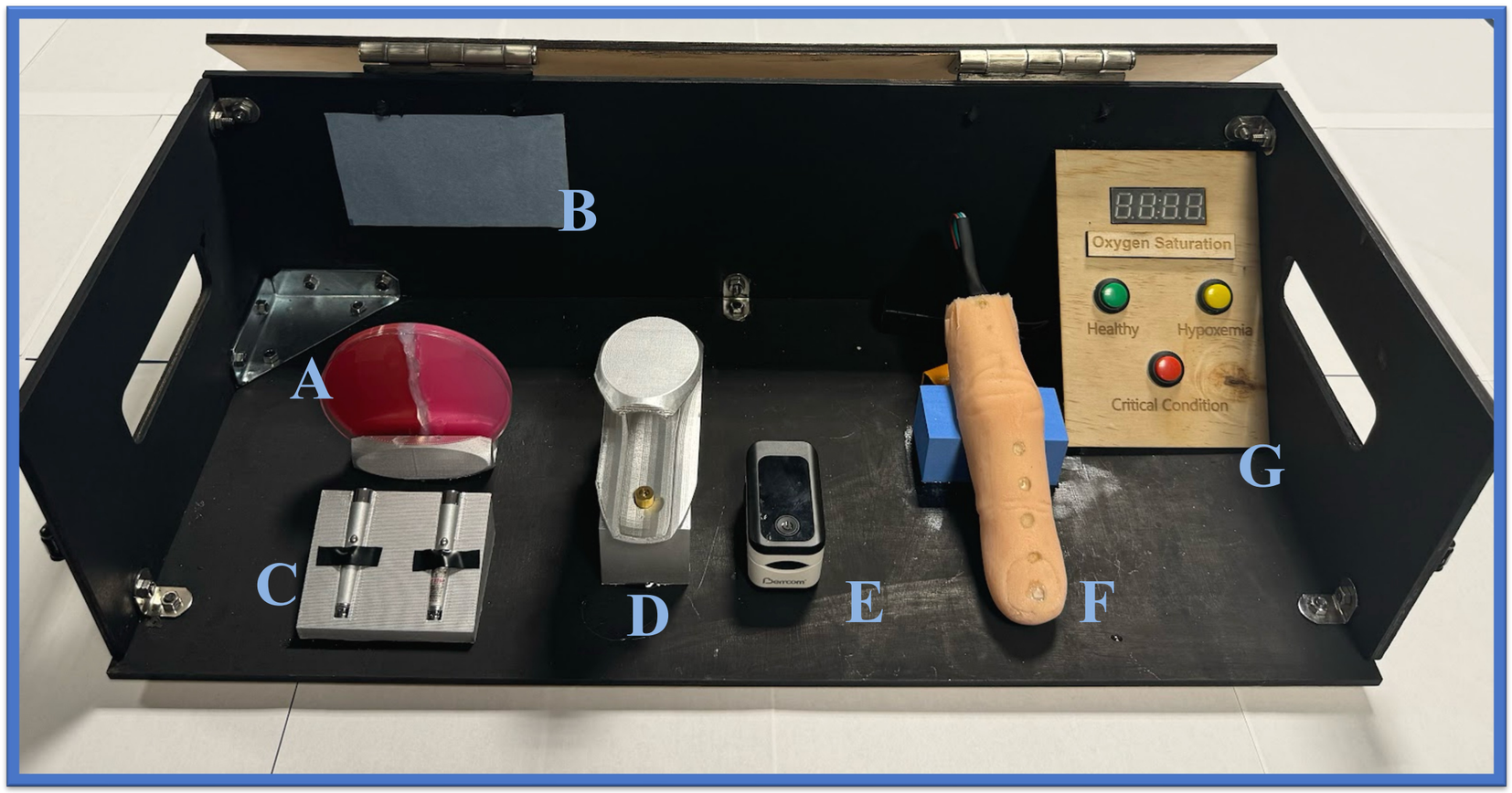
Background and Motivation

Pulse oximeters are essential medical devices, yet their underlying principles are rarely demonstrated interactively in STEM education

⇾ There exists a gap in accessible educational models that explain light absorption, blood oxygenation, and pulse oximetry

⇾ Interactive models are needed to provide the necessary visualization, interactivity, and knowledge retention for effective learning

Final Design Solution



Design Blocks and Functionality

Light Absorption Model: Laser with dual red gel filters of varying darkness

⇾ Simulates how oxygenated vs. deoxygenated blood absorbs red light

Finger Model: LED screen with buttons and a light-up finger

⇾ Buttons show 3 health conditions with matching oxygen levels and purple/red light ratios

⇾ Demonstrates how oxygenated blood ratio relate to value on pulse oximeter

Fake Pulse Oximeter: 3D-printed with a push-activated laser

⇾Students press to see red light pass through their finger

⇾ Shows red light's ability to pass through tissue

TA Sheet: Laminated lesson plan for the student TA

Pulse Oximeter: Real device used during the lesson

⇾ Students hold breath and watch oxygen levels drop

⇾ Demonstrates lower oxygen intake affects blood O_2 levels

Design Objectives and Testing

Design Objective	Target Value	Test	Results
Light Transmission Visibility	Average rating ≥4/5 on visibility scale	Test laser brightness through both gels in different lighting conditions	**PASS** - Average rating of 5/5 with 100% of participants easily distinguishing between lights
Educational Effectiveness and Interactivity	Average post-lesson education and interactivity survey scores ≥4	Mock classroom lesson followed by a post-survey evaluating perceived educational value and interactivity of the lesson	**PASS** – Average rating of 4.5/5 on both education and interactivity post survey
Durability	No major damage after rough handling; rating ≥4/5	Subject to repeated use and minor impacts	**PASS** - Maintained 100% functionality after testing; durability rated 5/5
Transportability	Average rating ≥4/5 on transport scale	Move model between locations and in vehicle	**PASS** - Average rating of 4.5/5; no component damage after transport trials
Ease of Use	All users successfully discovered and used every feature without prior instruction	Observe uninstructed users interacting with model	**PASS** - 100% of users independently identified and correctly utilized all features without any prior guidance

Diagram Key

A. Red gel filter
B. Light visibility viewing wall
C. Laser pointer housing
D. Fake pulse oximeter
E. Pulse Oximeter
F. Finger model
G. Interactive finger model panel

Conclusions and Next Steps

Conclusions:

⇾ The model effectively teaches pulse oximetry concepts to college students and is expected to yield similar results with middle and high school students

⇾ It promotes hands-on learning and student-centered exploration

⇾ It reinforces key principles through interactive and engaging activities

Moving forward, we hope to:

⇾ Have Dr. Madonna incorporate our interactive teaching model into her Duke Ignite teaching lesson

⇾ Conduct periodic checkups on teaching model for possible repairs

⇾ Explore options for expansion and building more complex models in EGR 102

References

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